A THREE-STATE POSTERIOR CONE AND RECONSTRUCTION ON FINITE-MEAN GALTON–WATSON TREES

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Abstract. We prove the sharp reconstruction threshold for the three-state ferromagnetic Potts channel on an observed Galton–Watson tree with an arbitrary finite-mean offspring law and $0 < \lambda < 1$. Reconstruction occurs exactly when $d\lambda^2 > 1$, and equality is nonreconstructible. The proof constructs a posterior-law cone defined by the low-degree invariant $H = y - x^2/2$. Two global pair inequalities, certified by exact nonnegative polynomial coefficients, preserve the cone and give a quadratic information saving. Averaging sequential branch fusion requires only the first offspring moment, so the theorem includes laws with infinite variance. It proves the three-state case of Formentin–Külske's general-offspring reconstruction conjecture and strengthens its strict subcritical conclusion to equality. We also give explicit critical decay bounds and almost-sure conclusions on survival. For comparison, we derive the corresponding four-state finite-mean statement from the existing four-state cone. The regular and Poisson threshold theorems and the four-state cone are prior results.

1. Introduction and main result

The reconstruction problem asks whether a noisy broadcast retains information about its root at arbitrarily large distances. For the symmetric q-state channel

$$K_\lambda(i,j) = \lambda \mathbf{1}_{\{i=j\}} + \frac{1-\lambda}{q}, \qquad i,j \in \{1,\dots,q\}, \tag{1}$$

the Kesten–Stigum parameter on a tree of mean offspring d is $d\lambda^2$. Reconstruction above this parameter's unit threshold follows from a second-moment argument. The difficult direction is to rule out nonlinear reconstruction at and below the threshold.

The contribution of this paper is an explicit three-state posterior cone that closes under the full broadcast recursion for every finite-mean offspring distribution. It gives a theorem for a specified published question: the three-state case of the general-offspring conjecture of Formentin and Külske [1, Conjecture 2.13, equation (45)]. Their condition is strictly below the Kesten–Stigum value; the argument here also includes equality.

Sly proved three-state sharpness at sufficiently large regular degree and non-sharpness for $q \geq 5$ [3]. Mossel, Sly and Sohn proved the three- and four-state Galton–Watson threshold at sufficiently large mean under uniform tail and tightness assumptions on the offspring family [2, Assumptions 1.10–1.11 and Theorem 1.12]. More recent manuscripts give the three-state symmetric and four-state ferromagnetic thresholds on every regular and observed Poisson tree [4, 5].

Our three-state construction uses the expectation condition $\mathbb{E}(y - x^2/2) \geq 0$, where x and y are the quadratic and cubic centered posterior invariants. Exact global pair inequalities preserve that condition and provide a strict quadratic information saving. The resulting offspring calculation depends on $\mathbb{E}(D-1)_+$, which is finite whenever $\mathbb{E}D$ is finite. In particular, it does not use a factorial

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second moment or an exponential offspring tail. The three-state pair certificate and its complete domain argument appear in Section 3 and Appendix A.

The method of using a cone of posterior laws, permutation averaging, and strict information savings is motivated by the four-state argument of [5]. We record its general-offspring consequence with full attribution. The new ingredient developed here is the particular three-state invariant and its global pair certificates; the four-state inequalities are imported from that earlier work. The finite-mean theorem extends the stated three-state regular/Poisson result of [4] and the large-mean, controlled-tail theorem of [2].

Let D be a nonnegative integer-valued random variable with finite mean d, and let T be a Galton–Watson tree with iid offspring D. The tree is independent of the spin process. The root spin σ_ρ is uniform on $\{1,\dots,q\}$; conditional on a parent spin, its children are generated independently through (1). The observer knows T and the spins on level $L_n(T)$. Put

$$p_n = \mathbb{P}(\sigma_\rho = \cdot \mid T, \sigma_{L_n(T)}), \qquad a_n(T) = \mathbb{E}_\sigma[\mathrm{TV}(p_n, u_q) \mid T], \qquad a_n = \mathbb{E}_T a_n(T), \tag{2}$$

where $u_q = (1/q,\dots,1/q)$. Reconstruction means that the corresponding limiting advantage is positive. For a fixed locally finite tree, $a_n(T)$ is nonincreasing by data processing, so this limit exists.

Theorem 1.1 (Three-state finite-mean reconstruction threshold)**.** *Let $q = 3$ and $0 < \lambda < 1$. For symmetric Potts broadcasting on the observed Galton–Watson tree described above:*

(1) If $d\lambda^2 \le 1$, then $a_n(T) \to 0$ for almost every tree.
(2) If $d\lambda^2 > 1$, then $\lim_n a_n(T) > 0$ for almost every surviving tree, and the limit is zero on extinction. In particular $\lim_n a_n > 0$.

Thus annealed reconstruction occurs if and only if $d\lambda^2 > 1$. The result requires only $\mathbb{E}D < \infty$ and includes every regular d-ary tree with integer $d \ge 2$.

Corollary 1.2 (The three-state general-offspring conjecture)**.** *For the three-state model, the non-reconstruction conclusion of Formentin–Külske* [1, Conjecture 2.13] *holds for every finite-mean offspring law. It remains valid at equality. In their temperature notation, with $\vartheta = \tanh\beta$ and $0 < \beta < \infty$, almost-sure nonreconstruction holds whenever*

$$d\left(\frac{2\vartheta}{3-\vartheta}\right)^2 \le 1.$$

Proof. The nontrivial eigenvalue of their transition matrix is

$$\lambda = \frac{e^{2\beta}-1}{e^{2\beta}+q-1} = \frac{2\vartheta}{q-(q-2)\vartheta}.$$

For $q = 3$, substitute this into Theorem 1.1. Their conjecture assumes strict inequality; the equality assertion follows from the critical estimate proved here. □

Corollary 1.3 (Four-state consequence of the prior cone)**.** *The conclusions of Theorem 1.1 also hold for $q = 4$. This is a consequence of the four-state posterior cone of* [5, Proposition 3.2] *through the first-moment argument in Section 5.*

1.1. **Offspring laws covered by the extension.** The distinction from a regular or Poisson tree is substantive. For example, the bounded offspring law

$$\mathbb{P}(D=1) = \frac{99}{100}, \qquad \mathbb{P}(D=101) = \frac{1}{100}$$

has $d = 2$ and $\mathbb{E}[D(D-1)]/d^2 = 101/4$. It is neither regular nor Poisson. As a second example, let

$$\mathbb{P}(D=k) = \frac{1}{\zeta(3)k^3}, \qquad k = 1,2,\dots,$$

where ζ is the Riemann zeta function. Then $d = \zeta(2)/\zeta(3) > 1$ but $\mathbb{E}D^2 = \infty$. Theorem 1.1 applies to both laws at every $0 < \lambda < 1$, including their exact critical values. The latter example lies outside the exponential-tail assumptions in [2].

At the three-state critical point, the proof gives the explicit bound

$$a_n \le \big(2(1+cn)\big)^{-1/2}, \qquad c = \frac{\lambda^4}{1000}\big(d - \mathbb{P}(D \ge 1)\big) > 0.$$

The constant is a sufficient certificate constant, not an asserted optimal rate constant. Above the threshold, a bounded pruning argument supplies reconstruction using only the first moment of the original offspring law.

1.2. **Scope and endpoints.** The exclusion of $\lambda = 1$ from the finite-noise theorem is substantive only at the deterministic unary tree: $D \equiv 1, \lambda = 1$ reconstructs perfectly although $d\lambda^2 = 1$. At $\lambda = 1$ and $d > 1$ reconstruction is perfect on survival; nondegenerate critical and subcritical Galton–Watson trees become extinct almost surely. The case $\lambda = 0$ is immediate. All results here concern the ferromagnetic regime; no antiferromagnetic conclusion is obtained by the cone argument.

2. Posterior experiments and Bayes products

An experiment with uniform input $X \in \{1, \dots, q\}$ induces a marginal law μ of the posterior vector $p = \mathbb{P}(X = \cdot \mid Y)$ in the simplex Δ_q. Its barycenter is u_q. We call the law symmetric if it is invariant under every permutation of the input labels.

Lemma 2.1 (Bayes product)**.** *Suppose two experiments are independent conditional on the common uniform input, and their marginal posterior laws are μ and ν. For independent $p \sim \mu$ and $r \sim \nu$, define*

$$U_i = qp_i r_i, \qquad z = \sum_i U_i, \qquad P_i = U_i/z \quad (z > 0).$$

For every bounded measurable f,

$$\mathbb{E}_{\mu\star\nu} f(P) = \mathbb{E}_{\mu\otimes\nu}[z f(P)], \tag{3}$$

with zero contribution when $z = 0$. If the laws are symmetric and f is permutation invariant, the integrand may be averaged over every relative permutation of r.

Proof. With respect to each observation's marginal law, its density conditional on $X = i$ is qp_i or qr_i. Conditional independence gives joint marginal density $\frac{1}{q}\sum_i (qp_i)(qr_i) = z$ relative to the product of the two marginals. Bayes' formula gives the displayed posterior and (3). The marginal posterior barycenters imply $\mathbb{E}z = 1$. If a normalizer is zero, that pair has zero probability under the joint observation law. Symmetry permits averaging the second marginal over relative permutations without changing its distribution. □

It is essential that independence in (3) is under the reference product measure. The actual joint observation measure contains the factor z.

Lemma 2.2 (Edges, mixtures, and additional observations)**.** *The following operations on posterior experiments have the stated properties.*

(1) Passing a child experiment through one channel edge maps its posterior to

$$p \longmapsto u_q + \lambda(p - u_q). \tag{4}$$

(2) An observed mixing label independent of the input mixes the corresponding posterior laws.

(3) If Φ is convex on Δ_q, observing additional information cannot decrease $\mathbb{E}\Phi(p)$.

Proof. The first assertion follows by applying the symmetric transition matrix to the child posterior; the uniform child marginal equals the uniform parent marginal. The second is conditioning on the observed label. For the third, the posterior before the added observation is the conditional expectation of the posterior afterward. Conditional Jensen gives the claim. □

3. An explicit three-state posterior cone

For $p \in \Delta_3$, let $v_i = 3p_i - 1$ and define

$$x(p) = \frac{1}{6}\sum_i v_i^2 = \frac{3\sum_i p_i^2 - 1}{2}, \qquad y(p) = \frac{1}{6}\sum_i v_i^3. \tag{5}$$

The functions used throughout this section are

$$\boxed{H(p) = y(p) - \tfrac{1}{2}x(p)^2, \qquad F(p) = \tfrac{3}{2}x(p) - y(p), \qquad G(p) = 1 - x(p).} \tag{6}$$

Lemma 3.1 (Scalar properties)**.** *For all $p \in \Delta_3$,*

$$0 \le x(p) \le 1, \qquad |y(p)| \le x(p)^{3/2} \le x(p), \qquad F(p) \ge \tfrac{1}{2}x(p) \ge 0, \qquad G(p) \ge 0.$$

The revealing posterior has $H = 1/2$, and the uniform posterior has $H = 0$. Under (4),

$$H(u_3 + \lambda(p - u_3)) = \lambda^3 H(p) + \tfrac{1}{2}\lambda^3(1-\lambda)x(p)^2 \ge \lambda^3 H(p) \quad (0 \le \lambda \le 1). \tag{7}$$

Proof. Set $\omega = e^{2\pi i/3}$ and $\zeta = p_1 + \omega p_2 + \omega^2 p_3$. Direct expansion gives $x = |\zeta|^2$ and $y = \mathrm{Re}(\zeta^3)$. Since $|\zeta| \le 1$, all bounds follow. At a point mass $x = y = 1$, whereas both vanish at u_3. Equation (4) multiplies ζ by λ, so x and y scale by λ^2 and λ^3. Substitution yields (7). □

For $p, r \in \Delta_3$ and $\pi \in S_3$, write

$$z_\pi = 3\sum_i p_i r_{\pi(i)}, \qquad P_i^\pi = \frac{3p_i r_{\pi(i)}}{z_\pi} \quad (z_\pi > 0), \qquad [zf(P)] = \frac{1}{6}\sum_{\pi \in S_3} z_\pi f(P^\pi). \tag{8}$$

Terms with $z_\pi = 0$ are zero.

Proposition 3.2 (Global three-state pair inequalities)**.** *With $\varepsilon = 1/1000$, every $p, r \in \Delta_3$ satisfy*

$$[zx(P)] \le x(p) + x(r) - H(p)F(r) - F(p)H(r) - \varepsilon x(p)x(r), \tag{9}$$

$$[zH(P)] \ge H(p)G(r) + G(p)H(r) + \varepsilon x(p)x(r). \tag{10}$$

Proof. We give the exact polynomial certificate and its domain argument. The independent integer implementation and complete verification data are described in Appendix A.

The expressions are invariant under separate permutations of p and r. A sorted vector $p_1 \ge p_2 \ge p_3$ is a convex combination of

$$e = (1,0,0), \qquad h = (1/2,1/2,0), \qquad u = (1/3,1/3,1/3),$$

with coefficients $p_1 - p_2$, $2(p_2 - p_3)$, and $3p_3$. Thus it suffices to work on the product of these two ordered chambers, in nonnegative barycentric triples each summing to one.

For a fixed permutation define

$$U_i = 3p_i r_{\pi(i)}, \qquad Z_\pi = \sum_i U_i, \qquad V_i = 3U_i - Z_\pi,$$

$$X_\pi = \tfrac{1}{6}\sum_i V_i^2, \qquad Y_\pi = \tfrac{1}{6}\sum_i V_i^3, \qquad N_\pi = Y_\pi Z_\pi - \tfrac{1}{2}X_\pi^2.$$

Direct substitution gives $Z_\pi x(P^\pi) = X_\pi/Z_\pi$ and $Z_\pi H(P^\pi) = N_\pi/Z_\pi^3$. Let

$$K_A = x(p) + x(r) - H(p)F(r) - F(p)H(r) - \varepsilon x(p)x(r),$$
$$K_B = -H(p)G(r) - G(p)H(r) - \varepsilon x(p)x(r).$$

The denominator-cleared targets are

$$T_A = 6K_A \prod_\pi Z_\pi - \sum_\pi X_\pi \prod_{\tau\neq\pi} Z_\tau, \tag{11}$$

$$T_B = 6K_B \prod_\pi Z_\pi^3 + \sum_\pi N_\pi \prod_{\tau\neq\pi} Z_\tau^3. \tag{12}$$

Homogenize each coordinate block to degree 10 for T_A and degree 22 for T_B. Multiplication by powers of the coordinate sums leaves values unchanged on the simplex. The supplied exact integer programs produce the following exhaustive coefficient counts:

Target	Total	Positive	Zero	Negative
T_A	4,356	3,873	483	0
T_B	76,176	71,627	4,549	0

Every coefficient is rational with positive denominator. Nonnegative coefficients prove $T_A, T_B \geq 0$ for all nonnegative barycentric coordinates. This is a global certificate on the initial chamber; no subdivision or sampled extrapolation is required. A separately written sparse six-variable implementation regenerates every coefficient and agrees exactly after the explicitly documented positive integer scaling.

For strictly positive p, r, all Z_π are positive, so division recovers (9)–(10). On the boundary, x and H are bounded on the compact simplex. Hence $Z_\pi f(P^\pi) \to 0$ whenever $Z_\pi \to 0$. Approximating each boundary pair by an interior pair proves the stated inequalities everywhere. □

Definition 3.3. Let $\mathcal{C}_3$ be the symmetric posterior laws μ such that $\mathbb{E}_\mu H \geq 0$.

The condition is imposed on an expectation, not on the support of the law. For example, $H(1/2, 1/2, 0) = -5/32$; such a posterior value is not excluded from an experiment whose average H is nonnegative.

Proposition 3.4 (Closure and strict subadditivity)**.** *The class $\mathcal{C}_3$ contains both full revelation and the uninformative experiment. It is preserved by positive channel steps, observed input-independent mixtures, and conditionally independent products. If $\mu, \nu \in \mathcal{C}_3$ and $m(\mu) = \mathbb{E}_\mu x$, then*

$$m(\mu \star \nu) \leq m(\mu) + m(\nu) - \varepsilon m(\mu)m(\nu). \tag{13}$$

Proof. The initial laws and edge closure follow from Lemma 3.1. Mixture closure follows by linearity and symmetry. For products, integrate (10) against $\mu \otimes \nu$ and use Lemma 2.1:

$$\mathbb{E}_{\mu\star\nu} H \geq (\mathbb{E}_\mu H)(\mathbb{E}_\nu G) + (\mathbb{E}_\mu G)(\mathbb{E}_\nu H) + \varepsilon(\mathbb{E}_\mu x)(\mathbb{E}_\nu x) \geq 0.$$

Product experiments remain symmetric. Integrating (9), the two products involving H and F are nonnegative because $F \geq 0$ and both H expectations are nonnegative. Dropping them proves (13). □

4. Four states: a consequence of the prior posterior cone

We use the four-state posterior cone and global pair inequalities of [5] and derive their consequences for arbitrary finite-mean offspring. We state the inequalities and describe their finite certificates explicitly.

For a posterior $p \in \Delta_4$, let

$$\zeta = Lp, \qquad L = \begin{pmatrix} 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}.$$

Define

$$A = \zeta_1^2 + \zeta_2^2 + \zeta_3^2 = 4\sum_{i=1}^{4} p_i^2 - 1, \qquad B = \zeta_1\zeta_2\zeta_3, \qquad C = \zeta_1^2\zeta_2^2 + \zeta_1^2\zeta_3^2 + \zeta_2^2\zeta_3^2. \tag{14}$$

In particular $0 \le A \le 3$. The three polynomials needed below are

$$H_4 = B - \frac{1}{10}A^2 - \frac{3}{20}C + \frac{3}{10}AB - \frac{1}{50}A^3 + \frac{7}{100}AC - \frac{51}{100}B^2, \tag{15}$$

$$F_4 = 4A - 3B + \frac{7}{10}A^2 - 4C - \frac{4}{5}AB, \tag{16}$$

$$G_4 = 1 - \frac{161}{100}A - \frac{31}{10}B + \frac{773}{1000}A^2 + 4C - \frac{199}{50}AB. \tag{17}$$

We regard these both as functions of p and as polynomials in $\zeta = Lp$; the argument will indicate which representation is used.

4.1. **The four-state inequalities.** For $p, r \in \Delta_4$ and $\pi \in S_4$, write

$$U_i^\pi = 4p_i r_{\pi(i)}, \qquad z_\pi = \sum_{i=1}^{4} U_i^\pi, \qquad P_i^\pi = \frac{U_i^\pi}{z_\pi} \quad (z_\pi > 0).$$

For a bounded function f on Δ_4, use the notation

$$[zf(P)]_4(p, r) = \frac{1}{24}\sum_{\pi \in S_4} z_\pi f(P^\pi),$$

where a summand with $z_\pi = 0$ is defined to be zero. Thus the brackets average over all twenty-four relative color permutations.

Proposition 4.1 (Four-state polynomial inequalities, [5])**.** *The polynomials* (15)–(17) *satisfy*

$$H_4 \ge -\frac{1}{2}, \qquad F_4 \ge 0, \qquad G_4 \ge 0 \quad \text{on } \Delta_4.$$

For $0 \le t \le 1$ *and* $\zeta \in L\Delta_4$*,*

$$H_4(t\zeta) \ge t^3 H_4(\zeta). \tag{18}$$

For every $p, r \in \Delta_4$*,*

$$[zA(P)]_4(p, r) \le A(p) + A(r) - H_4(p)F_4(r) - F_4(p)H_4(r) - \frac{A(p)A(r)(A(p) + A(r))}{1000}, \tag{19}$$

$$[zH_4(P)]_4(p, r) \ge H_4(p)G_4(r) + G_4(p)H_4(r). \tag{20}$$

4.2. **Finite certificates for the four-state inequalities.** We summarize the exact verification in [5] to specify precisely the four-state computational ingredient. Permuting the four probabilities acts on Lp by a signed permutation of three coordinates, with the product of the three signs equal to one. Consequently A, B, C and H_4, F_4, G_4 are invariant. The averaged pair expressions are invariant under separate permutations of p and r.

It therefore suffices to work on the ordered simplex

$$\mathcal{S}_4 = \{p \in \Delta_4 : p_1 \ge p_2 \ge p_3 \ge p_4\},$$

whose vertices are

$$e_1, \quad \frac{e_1+e_2}{2}, \quad \frac{e_1+e_2+e_3}{3}, \quad u_4.$$

Indeed its barycentric coordinates are $p_1-p_2, 2(p_2-p_3), 3(p_3-p_4), 4p_4$. On a simplex, a homogeneous polynomial with nonnegative ordinary monomial coefficients is nonnegative. Homogenization by powers of the sum of the barycentric coordinates does not alter values on the simplex. This gives the following exact scalar certificates; the displayed minimum refers to ordinary homogeneous coefficients.

Polynomial	Degree	Coefficients	Minimum
$H_4 + 1/2$	6	84	19/50
F_4	5	56	0
G_4	14	680	87/1000
$(3-\mathcal{D})H_4$	6	84	0

Here $\mathcal{D} = \sum_{j=1}^{3} \zeta_j \partial_{\zeta_j}$ is the Euler operator. Its scalar certificate implies (18), because

$$\frac{d}{dt}(t^{-3}H_4(t\zeta)) = t^{-4}(\mathcal{D}-3)H_4(t\zeta) \le 0 \qquad (0 < t \le 1),$$

and $H_4(0) = 0$ handles $t = 0$.

For the pair inequalities, define

$$\begin{aligned} D_0(p,r) &= \frac{A(p)A(r)(A(p)+A(r))}{1000}, \\ K_a &= 24\big(A(p)+A(r) - H_4(p)F_4(r) - F_4(p)H_4(r) - D_0(p,r)\big), \\ K_b &= -24\big(H_4(p)G_4(r) + G_4(p)H_4(r)\big). \end{aligned}$$

The two targets have the common form

$$\mathcal{R} = K + \sum_{\pi \in S_4} \frac{N_\pi}{z_\pi^k}, \tag{21}$$

with the following data:

Target	k	N_π	m
a	1	$-\|LU^\pi\|_2^2$	3
b	5	$z_\pi^6 H_4(U^\pi/z_\pi)$	1/2

The numerator in the second row is polynomial: each degree-s term of H_4 is multiplied by z_π^{6-s} after substitution. The two values of $\mathcal{R}$ are exactly twenty-four times the respective nonnegative differences in (19) and (20).

The verifier covers $\mathcal{S}_4 \times \mathcal{S}_4$ by products of rational subsimplices. On each cell it applies one of two sufficient, exact coefficient tests.

The first test uses

$$t_k(z) = \sum_{j=0}^{5} \binom{-k}{j} (z-1)^j \le z^{-k} \qquad (z > 0).$$

Taylor's theorem proves the inequality on the entire positive half-line: the sixth derivative of z^{-k} is positive, and the remainder contains the nonnegative factor $(z-1)^6$. The scalar bounds above give $N_\pi + m z_\pi^{k+1} \ge 0$. Hence

$$\frac{N_\pi}{z_\pi^k} \ge (N_\pi + m z_\pi^{k+1}) t_k(z_\pi) - m z_\pi.$$

Nonnegative homogeneous coefficients of the sum of these polynomial lower bounds and K certify the cell. The nonnegative multiplier is necessary; one cannot multiply a lower bound by an uncontrolled signed numerator.

For the second test, choose rational bihomogeneous polynomials J_π of bidegree $(3,3)$. With $W_j(X,Y) = (\sum_i X_i)^j (\sum_i Y_i)^j$, form

$$O_\pi = W_2 N_\pi - z_\pi^k J_\pi, \qquad Q_\pi = z_\pi^k W_3.$$

These are both of bidegree $(k+3, k+3)$, and Q_π has nonnegative coefficients. The checker chooses a rational $c_\pi \geq 0$ such that every coefficient of $O_\pi + c_\pi Q_\pi$ is nonnegative. A negative coefficient of O_π at a zero coefficient of Q_π makes this test fail. Otherwise the required correction is obtained by rounding each necessary coefficient ratio upward to a multiple of 2^{-32} and taking the largest correction. On the interior of the cell this proves

$$\frac{N_\pi}{z_\pi^k} = J_\pi - c_\pi + \frac{O_\pi + c_\pi Q_\pi}{z_\pi^k} \geq J_\pi - c_\pi.$$

The cell is accepted when the homogeneous coefficients of $K + \sum_\pi (J_\pi - c_\pi)$ are all nonnegative, after degree elevation. The polynomials J_π in the source are obtained by rational interpolation on the degree-three simplex lattice. Their interpolation accuracy is not assumed: the full remainder is certified by the displayed identity.

Unaccepted cells are bisected at an edge midpoint. If vertex i is replaced by the midpoint of vertices i, j, the barycentric substitution is

$$X_i = X_i'/2, \qquad X_j = X_j' + X_i'/2.$$

The binomial expansion gives exact rational coefficients for the child cell. The two choices of the replaced endpoint cover the original simplex, corresponding to $X_i \leq X_j$ and $X_j \leq X_i$. Repeating this construction therefore proves coverage, rather than merely testing a set of posterior points.

The retained full-verification records for the integer verifier of [5], run with assertions enabled on both targets, give the following complete traversals. Both terminated with no unresolved cells:

Target	Visited cells	Maximum depth	Taylor leaves	Remainder leaves
a	4,479	19	166	2,074
b	1,935	20	97	871

The scalar coefficients and the twenty-four induced coordinate actions were also checked independently by rational arithmetic. All sign decisions in the certificates use integer numerators and positive exact denominators.

Finally, denominator clearing and the lower bounds above are used first on cell interiors, where every $z_\pi > 0$. The weighted terms $z_\pi A(P^\pi)$ and $z_\pi H_4(P^\pi)$ converge to zero when $z_\pi \to 0$, since A, H_4 are bounded on Δ_4. They therefore have continuous extensions to every boundary point. This proves the boundary cases without dividing by a vanishing denominator. The original source contains the complete verifier and its deterministic subdivision rule.

4.3. **Closure of the four-state cone.** Let $\mathcal{C}_4$ consist of the permutation-symmetric posterior laws μ satisfying $\mathbb{E}_\mu H_4 \geq 0$. The condition is imposed on the law, not on each posterior in its support. Full revelation belongs to the cone because $H_4(e_i) = 13/100$; the uninformative experiment belongs because $H_4(u_4) = 0$.

Lemma 4.2 (Four-state experiment closure)**.** *The class $\mathcal{C}_4$ is preserved by a channel edge with $0 \leq \lambda \leq 1$, by conditionally independent products, and by observed mixtures with an input-independent mixing label. If $\mu, \nu \in \mathcal{C}_4$, and $a = \mathbb{E}_\mu A$, $b = \mathbb{E}_\nu A$, then*

$$\mathbb{E}_{\mu \star \nu} A \leq a + b - \frac{(\mathbb{E}_\mu A^2) b + a(\mathbb{E}_\nu A^2)}{1000} \leq a + b - \frac{ab(a+b)}{1000}. \tag{22}$$

Proof. An edge sends p to $u_4 + \lambda(p - u_4)$, and hence sends Lp to λLp. Equation (18) preserves the expectation condition, and A is multiplied by λ^2.

For the product of two experiments, the expectation of a posterior functional f is

$$\mathbb{E}_{\mu\star\nu} f = \mathbb{E}_{\mu\otimes\nu}[z f(P)].$$

The laws in the reference measure $\mu \otimes \nu$ are independent; the factor z gives the actual joint observation law. By permutation symmetry, averaging the integrand over the relative permutations does not change the expectation. Integrating (20) therefore gives

$$\mathbb{E}_{\mu\star\nu} H_4 \geq (\mathbb{E}_\mu H_4)(\mathbb{E}_\nu G_4) + (\mathbb{E}_\mu G_4)(\mathbb{E}_\nu H_4) \geq 0.$$

The product is symmetric as well. Integrating (19) and discarding the nonnegative products involving F_4 gives the first inequality in (22). Jensen's inequality gives the second. Observed input-independent mixtures preserve both symmetry and the expectation condition by linearity. □

5. Nonreconstruction for arbitrary finite-mean offspring

Let D be a nonnegative, almost surely finite integer-valued random variable with $d = \mathbb{E}D < \infty$, and let T be the associated Galton–Watson tree. The tree and all auxiliary child orderings are independent of the spins. The observer knows the tree and the spins at generation n. Knowing the continuation of the tree beyond that generation adds no spin information conditional on the observed finite prefix.

For $q = 3, 4$, write p_n for the posterior of the uniformly distributed root spin. Use the quadratic statistics

$$J_3(p) = \frac{3\sum_{i=1}^3 p_i^2 - 1}{2}, \qquad J_4(p) = 4\sum_{i=1}^4 p_i^2 - 1, \qquad m_n^{(q)} = \mathbb{E}J_q(p_n).$$

Thus $J_3 = x$ in the three-state section and $J_4 = A$ in the preceding section. Initially $m_0^{(3)} = 1$ and $m_0^{(4)} = 3$. Set

$$\rho = d\lambda^2, \qquad \delta = \mathbb{E}(D-1)_+ = d - \mathbb{P}(D \geq 1). \tag{23}$$

Only the first offspring moment is involved in δ.

Proposition 5.1 (Quadratic and cubic drift)**.** *For $0 \leq \lambda \leq 1$ and every $n \geq 0$,*

$$m_{n+1}^{(3)} \leq \rho m_n^{(3)} - c_3\big(m_n^{(3)}\big)^2, \qquad c_3 = \frac{\delta\lambda^4}{1000}, \tag{24}$$

$$m_{n+1}^{(4)} \leq \rho m_n^{(4)} - c_4\big(m_n^{(4)}\big)^3, \qquad c_4 = \frac{\delta\lambda^6}{500}. \tag{25}$$

Proof. At depth zero the fully revealing experiment belongs to the appropriate cone. Edge attenuation, conditionally independent products, and observed offspring mixtures show inductively that every finite-depth experiment belongs to that cone. These closures also apply to every partial product of incoming branch experiments.

Condition on $D = k$ and put $b = \lambda^2 m_n^{(q)}$. Every child branch has mean statistic b, since one channel edge multiplies the quadratic statistic by λ^2. If $j \geq 1$ branches have already been observed, their partial product has mean statistic at least b. To see this, let $P^{(j)}$ be the posterior after those j branches and $P^{(1)}$ the posterior after the first branch. The tower property gives

$$P^{(1)} = \mathbb{E}[P^{(j)} \mid \text{first branch observation}],$$

and conditional Jensen applies because both J_3 and J_4 are convex.

The three-state pair inequality proved earlier gives, for a partial product with mean $a \geq b$ and one further branch,

$$m(\text{new product}) \leq a + b - \frac{ab}{1000} \leq a + b - \frac{b^2}{1000}.$$

Applying this at each of the $k-1$ fusions, and including $k=0,1$, yields

$$\mathbb{E}[J_3(p_{n+1}) \mid D=k] \le kb - \frac{(k-1)_+ b^2}{1000}.$$

For four states, (22) instead gives a saving of at least

$$\frac{ab(a+b)}{1000} \ge \frac{b^3}{500}$$

at each fusion. Therefore

$$\mathbb{E}[J_4(p_{n+1}) \mid D=k] \le kb - \frac{(k-1)_+ b^3}{500}.$$

Averaging over the observed offspring count proves both recursions. The expectation is finite because $\mathbb{E}D < \infty$. □

Proposition 5.2 (Decay at and below the Kesten–Stigum value)**.** *Assume $0 < \lambda < 1$ and $\rho \le 1$. Let*

$$a_n^{(q)} = \mathbb{E}\,\mathrm{TV}(p_n, u_q).$$

If $\rho < 1$, then

$$m_n^{(3)} \le \rho^n, \qquad a_n^{(3)} \le \frac{\rho^{n/2}}{\sqrt{2}}, \tag{26}$$

$$m_n^{(4)} \le 3\rho^n, \qquad a_n^{(4)} \le \frac{\sqrt{3}}{2}\rho^{n/2}. \tag{27}$$

If $\rho = 1$, then $c_3, c_4 > 0$ and

$$m_n^{(3)} \le \frac{1}{1+c_3 n}, \qquad a_n^{(3)} \le \frac{1}{\sqrt{2(1+c_3 n)}}, \tag{28}$$

$$m_n^{(4)} \le (1/9 + 2c_4 n)^{-1/2}, \qquad a_n^{(4)} \le \frac{1}{2}(1/9 + 2c_4 n)^{-1/4}. \tag{29}$$

In both cases nonreconstruction holds for almost every realized tree.

Proof. Below criticality, omit the negative terms in (24)–(25) and iterate. The total-variation estimates follow from Cauchy–Schwarz:

$$\mathrm{TV}(p, u_3) \le \sqrt{J_3(p)/2}, \qquad \mathrm{TV}(p, u_4) \le \frac{1}{2}\sqrt{J_4(p)},$$

followed by Jensen's inequality.

At $\rho = 1$, the condition $\lambda < 1$ implies $d > 1$, and hence $\delta = d - \mathbb{P}(D \ge 1) > 0$. For three states, while $m_n = m_n^{(3)} > 0$,

$$\frac{1}{m_{n+1}} \ge \frac{1}{m_n(1 - c_3 m_n)} \ge \frac{1}{m_n} + c_3.$$

The denominator is positive: $m_n \le 1$ and $c_3 \le d\lambda^4/1000 \le 1/1000$ when $\rho \le 1$. If an iterate is zero, all subsequent iterates are zero and the stated bound is immediate. Otherwise iteration from $m_0 = 1$ proves (28).

For four states, write $m_n = m_n^{(4)}$. Since $m_n \le 3$ and $c_4 m_n^2 \le 9d\lambda^6/500 \le 9/500 < 1$, the analogous calculation gives

$$\frac{1}{m_{n+1}^2} \ge \frac{1}{m_n^2}(1 - c_4 m_n^2)^{-2} \ge \frac{1}{m_n^2} + 2c_4.$$

Iteration from $m_0 = 3$, with the same zero-iterate convention, proves (29).

For a fixed locally finite tree T, put

$$a_n^{(q)}(T) = \mathbb{E}[\mathrm{TV}(p_n, u_q) \mid T].$$

The root spin, level n, and level $n+1$ form a Markov chain, so $a_n^{(q)}(T)$ is nonincreasing by data processing. Its nonnegative limit therefore exists. The annealed estimates show that its expectation tends to zero. Bounded convergence yields

$$\mathbb{E}\big[\lim_{n\to\infty} a_n^{(q)}(T)\big] = 0,$$

so the limit is zero for almost every tree. The conclusion also holds almost surely conditional on survival whenever survival has positive probability. No independence assumption after conditioning on survival is needed. □

6. Reconstruction above the threshold

The argument in this section works for every finite number of states $q \geq 2$. It supplies the supercritical direction with only a finite first offspring moment.

Proposition 6.1 (Bounded pruning)**.** *Let $\mathbb{E}D = d < \infty$ and $0 < \lambda < 1$. If $d\lambda^2 > 1$, then the annealed reconstruction advantage is bounded away from zero uniformly in depth.*

Proof. By monotone convergence, choose $L < \infty$ such that

$$D_L = \min(D, L), \qquad \mu = \mathbb{E}D_L, \qquad \mu\lambda^2 > 1.$$

Use the original Ulam–Harris child indexing in the iid offspring construction and keep the first D_L children at every retained vertex. The indexing is fixed before the descendant offspring and spins are generated; in particular, it does not select children according to their subtree shapes. If the observed tree is regarded as unordered, adjoin independent uniform sibling orderings, which carry no spin information conditional on the tree. The retained subtree has iid bounded offspring D_L, and its depth-n spins are part of the original depth-n observation (with this independent auxiliary ordering).

Choose a channel eigenfunction ϕ with uniform mean zero and second moment one; for example,

$$\phi(1) = \sqrt{q-1}, \qquad \phi(i) = -\frac{1}{\sqrt{q-1}} \quad (2 \leq i \leq q).$$

It satisfies $\mathbb{E}[\phi(\sigma_{\mathrm{child}}) \mid \sigma_{\mathrm{parent}}] = \lambda\phi(\sigma_{\mathrm{parent}})$ and $|\phi| \leq B_q := \sqrt{q-1}$. On the retained tree define

$$Z_n = (\mu\lambda)^{-n} \sum_{|v|=n} \phi(\sigma_v).$$

The first-moment recursion gives

$$\mathbb{E}[Z_n \mid \sigma_{\mathrm{root}}] = \phi(\sigma_{\mathrm{root}}), \qquad \mathbb{E}[\phi(\sigma_{\mathrm{root}})Z_n] = 1. \tag{30}$$

Let $M_n = \mathbb{E}Z_n^2$ and $\alpha_L = \mathbb{E}[D_L(D_L-1)]/\mu^2$. Writing the depth-$(n+1)$ sum as a sum of child contributions gives

$$M_{n+1} = \frac{M_n}{\mu\lambda^2} + \alpha_L, \qquad M_0 = 1. \tag{31}$$

Indeed the diagonal terms have total expectation μM_n. For distinct children, conditional independence given the root and (30) give product expectation $\lambda^2\phi(\sigma_{\mathrm{root}})^2$, whose mean is λ^2. Dividing by $(\mu\lambda)^2$ proves (31). Since D_L is bounded and $r_L = (\mu\lambda^2)^{-1} < 1$,

$$M_n \leq C_L := 1 + \frac{\alpha_L}{1-r_L} < \infty \qquad \text{for all } n.$$

Let $\mathcal{F}_n$ contain the original tree, its level-n spins, and the auxiliary ordering. The statistic Z_n is $\mathcal{F}_n$-measurable. Put $\eta_n = \mathbb{E}[\phi(\sigma_{\mathrm{root}}) \mid \mathcal{F}_n]$. Cauchy–Schwarz and (30) imply

$$1 = \mathbb{E}[\eta_n Z_n] \le \left(\mathbb{E}\eta_n^2\, C_L\right)^{1/2}, \qquad \mathbb{E}\eta_n^2 \ge C_L^{-1}.$$

The ordering is independent of the root and spins conditional on the original observed tree. Consequently adjoining it does not change the root posterior. For that posterior,

$$|\eta_n| \le B_q, \qquad \mathrm{TV}(p_n, u_q) \ge \frac{|\eta_n|}{2B_q}.$$

Therefore

$$a_n^{(q)} \ge \frac{\mathbb{E}|\eta_n|}{2B_q} \ge \frac{\mathbb{E}\eta_n^2}{2B_q^2} \ge \frac{1}{2(q-1)C_L} > 0. \tag{32}$$

Only the truncated offspring variable was required to have a second moment. □

Lemma 6.2 (The reconstruction event on a Galton–Watson tree)**.** *For $0 < \lambda < 1$ and $d\lambda^2 > 1$, reconstruction holds for almost every surviving Galton–Watson tree and fails on the extinction event.*

Proof. First fix a deterministic locally finite rooted tree. Its root reconstructs if and only if at least one of its child subtrees reconstructs. If a child reconstructs, observing only that child's descendants gives the root posterior

$$u_q + \lambda(p_{\mathrm{child}} - u_q).$$

Its total-variation advantage is λ times the child's advantage, so remains positive in the limit.

Conversely, suppose all child subtrees are nonreconstructing. For a finite input alphabet, vanishing expected posterior total variation implies vanishing mutual information: the posterior relative entropy from the uniform prior is bounded and continuous on the compact simplex. Thus every child's boundary observation has vanishing mutual information with the child spin and, by data processing, with the root spin. There are only finitely many children. Their observations $Y_1, \dots, Y_k$ are independent conditional on the root, and hence

$$I(\sigma_{\mathrm{root}}; Y_1, \dots, Y_k) \le \sum_{j=1}^{k} I(\sigma_{\mathrm{root}}; Y_j) \longrightarrow 0.$$

Here the inequality follows from subadditivity of unconditional entropy and additivity of conditional entropy. Pinsker's inequality then implies nonreconstruction at the root.

Let r_* be the probability that a Galton–Watson tree reconstructs. The preceding equivalence and independence of child subtrees yield

$$r_* = 1 - \mathbb{E}(1 - r_*)^D.$$

Let ξ be the extinction probability. Since $d\lambda^2 > 1$ implies $d > 1$, the offspring generating function has exactly the two fixed points ξ and 1 in $[0, 1]$. For completeness, it is strictly convex on $(0, 1)$ because $\mathbb{P}(D \ge 2) > 0$; its derivative at 1 equals $d > 1$, and ξ is its least fixed point. These facts exclude any third fixed point.

The positive lower bound (32), together with bounded convergence for the fixed-tree advantage, implies $r_* > 0$. Thus $1 - r_* = \xi$. Reconstruction is impossible on an extinct tree, whose boundary is eventually empty. Consequently the reconstruction event agrees with survival up to a null set. □

7. The finite-mean dichotomy and endpoint conventions

Theorem 7.1 (Three and four states on finite-mean Galton–Watson trees)**.** *Let D be a nonnegative, almost surely finite integer-valued random variable with finite mean d. Consider the symmetric ferromagnetic q-state broadcast process on its observed Galton–Watson tree, with $q \in \{3, 4\}$ and $0 < \lambda < 1$. If $d\lambda^2 \le 1$, nonreconstruction holds for almost every tree. If $d\lambda^2 > 1$, reconstruction holds for almost every surviving tree and fails on extinct trees. In particular, annealed reconstruction occurs if and only if*

$$d\lambda^2 > 1.$$

No offspring moment beyond $\mathbb{E}D < \infty$ is assumed. At equality, the quantitative bounds are (28) *and* (29)*.*

Proof. Proposition 5.2 proves the assertion at and below equality. Proposition 6.1 and Lemma 6.2 prove the assertions above equality. □

Remark 7.2 (Regular trees and endpoints). Taking $D \equiv d$ gives every regular tree of integer degree $d \ge 2$. At $\lambda = 0$, all positive-depth observations are independent of the root. At $\lambda = 1$, every descendant copies the root, so reconstruction is perfect on survival and impossible after extinction. A subcritical or nondegenerate critical finite-mean Galton–Watson process dies out almost surely. The exceptional law $D \equiv 1$ instead gives an infinite noiseless line: at $\lambda = 1$ it reconstructs perfectly although $d\lambda^2 = 1$. A statement including all closed endpoints must record this exception. The argument concerns nonnegative channel eigenvalues; it makes no assertion about the antiferromagnetic case.

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Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The three-state argument, the finite-mean offspring deductions, the literature search, the verification code and the manuscript were produced by OpenAI AI systems under the author's direction. The four-state posterior-cone theorem is prior work and is attributed explicitly in the text. The computations were checked by exact certificate reruns and an independent reconstruction of all three-state polynomial coefficients. The author selected the problem and commissioned and directed the work; the author did not carry out the mathematics or the writing. Responsibility for any decision to submit or publish the manuscript rests with the author.

The present version adds a separate Python exact-arithmetic reconstruction of both three-state certificate polynomials. It checks all 80 532 coefficient positions without importing either earlier polynomial implementation or its coefficient tables. Separate AI-assisted reviews examined the first-moment drift, critical decay, pruning, and survival arguments. These checks are internal

mathematical and computational verification; they are not external peer review or a proof-assistant formalization.

Appendix A. Exact coefficient certificates and reproducibility

This appendix specifies the finite arithmetic certificates used in the proof. The three-state inequalities are verified by two different polynomial constructions. The four-state inequalities use the published certificate of [5], whose two complete traversals were rerun. All sign decisions use integer or rational arithmetic.

A.1. **The three-state residuals and their complete domain.** For $p \in \Delta_3$, retain the definitions

$$x(p) = \frac{1}{6}\sum_i (3p_i - 1)^2, \qquad y(p) = \frac{1}{6}\sum_i (3p_i - 1)^3, \qquad H = y - \frac{x^2}{2}, \quad F = \frac{3}{2}x - y, \quad G = 1 - x.$$

For $\pi \in S_3$, put

$$z_\pi = 3\sum_i p_i q_{\pi(i)}, \qquad r_{\pi,i} = \frac{3p_i q_{\pi(i)}}{z_\pi}, \qquad \epsilon = \frac{1}{1000}.$$

A term $z_\pi f(r_\pi)$ is defined as zero when $z_\pi = 0$. The two residuals certified below are

$$\begin{aligned} R_A(p,q) &= x(p) + x(q) - \frac{1}{6}\sum_{\pi\in S_3} z_\pi x(r_\pi) \\ &\quad - H(p)F(q) - F(p)H(q) - \epsilon x(p)x(q), \end{aligned} \tag{33}$$

$$\begin{aligned} R_B(p,q) &= \frac{1}{6}\sum_{\pi\in S_3} z_\pi H(r_\pi) \\ &\quad - H(p)G(q) - G(p)H(q) - \epsilon x(p)x(q). \end{aligned} \tag{34}$$

It suffices to work in the ordered chambers for p and q: all scalar functions are symmetric, and the fusion terms average over every relative permutation. Introduce nonnegative variables (a, b, c), (d, e, f), and let $S = a + b + c > 0$, $T = d + e + f > 0$. Set

$$p = \frac{(6a + 3b + 2c,\ 3b + 2c,\ 2c)}{6S}, \qquad q = \frac{(6d + 3e + 2f,\ 3e + 2f,\ 2f)}{6T}.$$

This covers the whole ordered simplex. Indeed, for $p_1 \geq p_2 \geq p_3$, the normalized coordinates

$$(a, b, c) = (p_1 - p_2,\ 2(p_2 - p_3),\ 3p_3)$$

are nonnegative and sum to one. Thus the certificate is not restricted to sampled posterior vectors or to a neighborhood of the uniform law.

A.2. **An explicit integer polynomial certificate.** Define

$$X = 4a^2 + 2ab + b^2, \qquad Y = 8a^3 + 6a^2 b - 3ab^2 - b^3,$$

and use primes for the same expressions in (d, e, f). Direct substitution gives

$$x(p) = \frac{X}{4S^2}, \qquad y(p) = \frac{Y}{8S^3}.$$

Writing

$$\widehat{H} = 4YS - X^2, \qquad \widehat{F} = 3XS - Y, \qquad \widehat{G} = 4S^2 - X,$$

we have

$$H(p) = \frac{\widehat{H}}{32S^4}, \qquad F(p) = \frac{\widehat{F}}{8S^3}, \qquad G(p) = \frac{\widehat{G}}{4S^2}.$$

The auxiliary signs used in the expectation argument follow, for example, from the explicit expansions

$$\widehat{F} = 4a^3 + 12a^2b + 12ab^2 + 4b^3 + 12a^2c + 6abc + 3b^2c,$$
$$\widehat{G} = 6ab + 8ac + 3b^2 + 8bc + 4c^2.$$

Let $P = (6a + 3b + 2c, 3b + 2c, 2c)$ and $Q = (6d + 3e + 2f, 3e + 2f, 2f)$. For each permutation define

$$U_{\pi,i} = P_i Q_{\pi(i)}, \quad Z_\pi = \sum_i U_{\pi,i}, \quad W_{\pi,i} = 3U_{\pi,i} - Z_\pi,$$

$$V_\pi = \sum_i W_{\pi,i}^2, \qquad J_\pi = 36 Z_\pi W_{\pi,1} W_{\pi,2} W_{\pi,3} - V_\pi^2.$$

Since $\sum_i W_{\pi,i} = 0$, the weighted fusion terms satisfy

$$z_\pi = \frac{Z_\pi}{12ST}, \qquad x(r_\pi) = \frac{V_\pi}{6Z_\pi^2}, \qquad H(r_\pi) = \frac{J_\pi}{72Z_\pi^4}.$$

In particular, the factor z_π from the posterior product law reduces the denominator powers to one and three.

Set $D_A = \prod_\pi Z_\pi$, $D_B = \prod_\pi Z_\pi^3$, and define

$$K_A = 8000(XS^2T^4 + X'S^4T^2) - 125(\widehat{H}\widehat{F}'T + \widehat{F}\widehat{H}'S) - 2XX'S^2T^2,$$
$$K_B = -125(\widehat{H}\widehat{G}'T^2 + \widehat{G}\widehat{H}'S^2) - XX'S^2T^2.$$

The following two polynomials have integer coefficients:

$$\mathcal{P}_A = 27K_AD_A - 2000S^3T^3 \sum_\pi V_\pi \prod_{\rho \neq \pi} Z_\rho, \tag{35}$$

$$\mathcal{P}_B = 81K_BD_B + 250S^3T^3 \sum_\pi J_\pi \prod_{\rho \neq \pi} Z_\rho^3. \tag{36}$$

Their exact relation to the desired residuals is

$$\mathcal{P}_A = 864000S^4T^4D_AR_A, \qquad \mathcal{P}_B = 1296000S^4T^4D_BR_B. \tag{37}$$

Expanding (35)–(36) in the six nonnegative barycentric variables gives the complete coefficient inventory:

Certificate	Bidegree	Positions	Positive	Zero	Negative
A	$(10, 10)$	4 356	3 873	483	0
B	$(22, 22)$	76 176	71 627	4 549	0

The number of positions is $\binom{n+2}{2}^2$ in bidegree (n, n). Every coefficient is checked; no subdivision is needed for these three-state polynomials.

Nonnegative coefficients imply $\mathcal{P}_A, \mathcal{P}_B \geq 0$. For interior probability vectors, $Z_\pi > 0$ for all π, so (37) proves $R_A, R_B \geq 0$. Boundary points are included by continuity: x and H are bounded on the probability simplex, hence $z_\pi x(r_\pi)$ and $z_\pi H(r_\pi)$ tend to zero when z_π tends to zero. Approximation by interior pairs therefore establishes (33)–(34) on the entire domain.

A.3. **Independent verification.** The source `analytic/q3_exact_pair_simple.cpp` constructs rational homogeneous polynomials directly from the definitions of the posterior invariants. Separately, `computation/q3_independent_coefficients.cpp` expands (35)–(36) in a sparse six-variable integer representation. Its construction does not use the first encoder's invariants, homogenization, tensor multiplication, or rational-polynomial operations. Only after constructing its polynomials does it call the first encoder to compare the complete coefficient arrays.

If $\mathcal{C}_A, \mathcal{C}_B$ denote the rational homogeneous outputs of the first encoder, every coefficient satisfies the exact identities

$$\mathcal{P}_A = 144000 \cdot 12^6\, \mathcal{C}_A, \qquad \mathcal{P}_B = 216000 \cdot 12^{18}\, \mathcal{C}_B.$$

The comparison uses integer cross-products with the stored denominators. Both constructions use arbitrary-precision integers. The independent implementation packs six exponents into five-bit fields and checks that each group's total degree is below 32; the maximum used is 22, so exponent addition cannot carry between fields.

Both implementations completed with assertions enabled and UndefinedBehaviorSanitizer. The clean certificates accept at their root, so floating-point edge-selection coordinates in the first encoder are never consulted. No sampled inequality or floating-point sign decision is used.

The supplementary files contain both implementations, the complete coefficient tables, and the verification output. The independent program regenerates every coefficient, including all zeros.

A.4. **A dependency-free exact reconstruction.** The supplementary program `computation/verify_q3_python.py` independently constructs the integer polynomials in (35)–(36) using Python's arbitrary-precision integers. Its sparse polynomial representation and multiplication routines do not import the earlier C++ encoders or any stored coefficient table. The program checks every homogeneous coefficient position, including zero entries. The full expansion reproduces the two inventories above: 3 873 positive and 483 zero entries for A, and 71 627 positive and 4 549 zero entries for B, with no negative coefficient.

Separate exact-rational substitutions compare the cleared polynomials with the direct posterior residuals in (33)–(34). These finite identity checks include boundary inputs; the global sign proof comes from the exhaustive nonnegative coefficient expansion and the domain argument above. The program, machine-readable result, and regenerated coefficient tables are included in the supplement.

A.5. **Four-state certificate.** The four-state pair inequalities and their certificate algorithm are those of [5]; the supplementary files include the unchanged verifier from the version linked in that reference.

The verifier works on products of ordered posterior simplices and sums over all 24 relative colour permutations. It certifies subdomains using either a Taylor lower bound for reciprocal powers with a nonnegative multiplier, or cubic interpolation with a rigorously bounded cleared polynomial remainder. Exact midpoint bisection covers a subdomain whenever neither certificate yet applies. A successful traversal certifies every resulting subdomain; see [5] for the inequalities and the justification of these two tests.

Both full traversals were executed with the published C++ source unchanged, exact integer/rational arithmetic, and assertions enabled:

Mode	Nodes	Max. depth	Interpolation leaves	Taylor leaves
A	4 479	19	2 074	166
B	1 935	20	871	97

Both exited with status zero. Their terminal lines were

```
done 4479 19 lp=2074 lt=166
done 1935 20 lp=871 lt=97
```

Complete output records are included with the supplementary code.

Additional checks verify the four homogeneous scalar polynomials' coefficient minima, all 24 symmetry actions, and the exact arithmetic, interpolation, and subdivision operations under UndefinedBehaviorSanitizer. The scalar coefficient minima are $19/50, 0, 87/1000, 0$ for $H + 1/2, F, G, (3 - \mathrm{Euler})H$,

respectively; these are coefficient minima, not asserted minima of the functions. The lightweight checks are supplemented by the two complete traversals reported above.

A.6. **Compact reproduction instructions.** The accompanying source archive preserves the cited files and their relative paths. The Python reconstruction needs only Python 3 and its standard library:

```
python3 computation/verify_q3_python.py
```

For the original C++ checks, requirements are Python 3, a C++17 compiler with the GCC-compatible headers used by the published verifier, and Boost multiprecision headers. Keep assertions enabled. If Boost is outside the compiler's default include path, set `CPLUS_INCLUDE_PATH` to its include directory. From the archive root:

```
cd computation
g++ -std=c++17 -O2 -UNDEBUG -fsanitize=undefined \
  -fno-sanitize-recover=all \
  ../analytic/q3_exact_pair_simple.cpp -o q3_check
./q3_check 0
./q3_check 1
g++ -std=c++17 -O2 -UNDEBUG -fsanitize=undefined \
  -fno-sanitize-recover=all \
  q3_independent_coefficients.cpp -o q3_independent
./q3_independent 0
./q3_independent 1
cd published_q4/verification
python3 checks/verify.py
g++ -std=c++17 -O3 -UNDEBUG \
  checks/certificate.cpp -o artifacts/certificate
./artifacts/certificate
./artifacts/certificate b
```

Every command must exit with status zero. The q3 outputs must reproduce the coefficient counts above, and the q4 executions must print the two stated completion lines. The probabilistic deductions from these algebraic inequalities are supplied in the main text.

Public repository and code availability

This paper and its reproducibility package are archived in the public Math Brain repository [6]; the paper-specific archive and PDF are given in [7]. The supplementary material contains the complete LaTeX source, both original three-state verification programs, a separate dependency-free Python verifier, complete coefficient tables, and the four-state verifier of [5], with instructions for reproducing the exact checks. The version of the four-state source is identified in the bibliography.

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